

Scaling Microservices for High Traffic Using Docker Compose and Nginx

```
Command Prompt
Volume Serial Number is AEE6-4A19

Directory of D:\scaling-demo

13-05-2026 18:17 <DIR>      .
13-05-2026 17:44 <DIR>      backend
13-05-2026 18:35          530 docker-compose.yml
13-05-2026 17:48 <DIR>      nginx
                1 File(s)      530 bytes
                3 Dir(s)  137,628,246,016 bytes free

D:\scaling-demo>type docker-compose.yml
services:
  backend:
    build: ./backend
    expose:
      - "8080"
    restart: unless-stopped
    healthcheck:
      test: ["CMD", "curl", "-f", "http://localhost:8080"]
      interval: 10s
      timeout: 3s
      retries: 5
      start_period: 10s

  frontend:
    image: nginx:latest
    container_name: frontendcont
    ports:
      - "80:80"
    depends_on:
      backend:
        condition: service_healthy
    restart: unless-stopped
    volumes:
      - ./nginx/nginx.conf:/etc/nginx/nginx.conf:ro
D:\scaling-demo>
```

```
D:\scaling-demo\backend>type dockerfile
FROM python:3.11-slim
WORKDIR /app
RUN apt-get update && apt-get install -y curl
COPY requirements.txt .
RUN pip install -r requirements.txt
COPY . .
CMD ["python", "app.py"]
D:\scaling-demo\backend>
```

```
localhost
localhost
Machine Learning Neural Networks ar... Density-based clust... Fisher LDA-Stanford Competitive Progra... CVB in Unsupervise... Red Hat Academy System Design Primer
Pretty-print
{
  "container": "4cf9eed432a8",
  "message": "Backend container is running"
}
```

Before Scaling

After Scaling:

```
Command Prompt

#12 exporting manifest sha256:341d000778c53071c6e991f275dc900872c320f5ed1b65a0dfcea8c7735985a7 done
#12 exporting config sha256:b5ca3c49a335b5762cabf274934f146f6f3f08ec918543aala77dc332f698c39 done
#12 exporting attestation manifest sha256:ef1f356ebe3e371e61c13b30579095e127ac6130264c02016e51172e08031b39 0.1s done
#12 exporting manifest list sha256:10f367f79ce82c642d6dcca1ea90ed101522b3b5871a0f5ea1b237d1ef905a48 0.0s done
#12 naming to docker.io/library/scaling-demo-backend:latest done
#12 unpacking to docker.io/library/scaling-demo-backend:latest 0.0s done
#12 DONE 0.2s

[+] up 6/6ing provenance for metadata file
✓Image scaling-demo-backend Built 6.5s
✓Network scaling-demo_default Created 0.1s
✓Container scaling-demo-backend-3 Created 0.5s
✓Container scaling-demo-backend-1 Created 0.5s
✓Container scaling-demo-backend-2 Created 0.5s
✓Container frontendcont Created 0.6s
Attaching to frontendcont, backend-1, backend-2, backend-3
backend-3 | * Serving Flask app 'app'
backend-3 | * Debug mode: off
backend-3 | WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI serv
er instead.
backend-3 | * Running on all addresses (0.0.0.0)

Container scaling-demo-backend-3 Waiting
Container scaling-demo-backend-1 Waiting
Container scaling-demo-backend-2 Waiting
backend-1 | * Serving Flask app 'app'

backend-3 | * Running on http://127.0.0.1:8080
backend-1 | * Debug mode: off
```

```
Command Prompt - docker c

backend-2 | WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI serv
er instead.
backend-2 | * Running on all addresses (0.0.0.0)
backend-2 | * Running on http://127.0.0.1:8080
backend-2 | * Running on http://172.24.0.4:8080
backend-2 | Press CTRL+C to quit
backend-3 | 127.0.0.1 - - [13/May/2026 13:07:06] "GET / HTTP/1.1" 200 -
backend-1 | 127.0.0.1 - - [13/May/2026 13:07:07] "GET / HTTP/1.1" 200 -
backend-2 | 127.0.0.1 - - [13/May/2026 13:07:08] "GET / HTTP/1.1" 200 -
Container scaling-demo-backend-3 Healthy
Container scaling-demo-backend-1 Healthy
Container scaling-demo-backend-2 Healthy
frontendcont | /docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
frontendcont | /docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
frontendcont | /docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
frontendcont | 10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
frontendcont | 10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
frontendcont | /docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
frontendcont | /docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
frontendcont | /docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
frontendcont | /docker-entrypoint.sh: Configuration complete; ready for start up
backend-2 | 172.24.0.5 - - [13/May/2026 13:07:14] "GET / HTTP/1.1" 200 -
frontendcont | 172.24.0.1 - - [13/May/2026 13:07:14 +0000] "GET / HTTP/1.1" 200 70 "-" "Mozilla/5.0 (Windows NT 10.0; W
in64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/148.0.0.0 Safari/537.36"
backend-3 | 127.0.0.1 - - [13/May/2026 13:07:16] "GET / HTTP/1.1" 200 -
backend-1 | 127.0.0.1 - - [13/May/2026 13:07:17] "GET / HTTP/1.1" 200 -
backend-2 | 127.0.0.1 - - [13/May/2026 13:07:18] "GET / HTTP/1.1" 200 -
backend-1 | 172.24.0.5 - - [13/May/2026 13:07:20] "GET / HTTP/1.1" 200 -
frontendcont | 172.24.0.1 - - [13/May/2026 13:07:20 +0000] "GET / HTTP/1.1" 200 70 "-" "Mozilla/5.0 (Windows NT 10.0; W
in64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/148.0.0.0 Safari/537.36"
```

```
Command Prompt - docker c x + v
in64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/148.0.0.0 Safari/537.36"
backend-1 | 172.24.0.5 - - [13/May/2026 13:08:08] "GET / HTTP/1.1" 200 -
backend-1 | 127.0.0.1 - - [13/May/2026 13:08:08] "GET / HTTP/1.1" 200 -
backend-2 | 127.0.0.1 - - [13/May/2026 13:08:09] "GET / HTTP/1.1" 200 -
backend-3 | 172.24.0.5 - - [13/May/2026 13:08:13] "GET / HTTP/1.1" 200 -
frontendcont | 172.24.0.1 - - [13/May/2026:13:08:13 +0000] "GET / HTTP/1.1" 200 70 "-" "Mozilla/5.0 (Windows NT 10.0; W
in64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/148.0.0.0 Safari/537.36"

backend-3 | 127.0.0.1 - - [13/May/2026 13:08:17] "GET / HTTP/1.1" 200 -
backend-1 | 127.0.0.1 - - [13/May/2026 13:08:18] "GET / HTTP/1.1" 200 -

backend-2 | 127.0.0.1 - - [13/May/2026 13:08:19] "GET / HTTP/1.1" 200 -

backend-3 | 127.0.0.1 - - [13/May/2026 13:08:27] "GET / HTTP/1.1" 200 -
backend-1 | 127.0.0.1 - - [13/May/2026 13:08:28] "GET / HTTP/1.1" 200 -

backend-2 | 127.0.0.1 - - [13/May/2026 13:08:29] "GET / HTTP/1.1" 200 -
backend-3 | 127.0.0.1 - - [13/May/2026 13:08:37] "GET / HTTP/1.1" 200 -

backend-1 | 127.0.0.1 - - [13/May/2026 13:08:38] "GET / HTTP/1.1" 200 -
backend-2 | 127.0.0.1 - - [13/May/2026 13:08:39] "GET / HTTP/1.1" 200 -
```

```
localhost x +
localhost
Machine Learning Neural Networks ar... Density-based clust... Fisher LDA-Stanford Competitive Progra... CV8 in Unsupervise... Red Hat Academy System Design Primer
Pretty-print
{
  "container": "5a3b6e2b68bd",
  "message": "Backend container is running"
}
```

```
localhost x +
localhost
Machine Learning Neural Networks ar... Density-based clust... Fisher LDA-Stanford Competitive Progra... CV8 in Unsupervise... Red Hat Academy System Design Primer
Pretty-print
{
  "container": "0fb436f4f46c",
  "message": "Backend container is running"
}
```

```
localhost x +
localhost
Machine Learning Neural Networks ar... Density-based clust... Fisher LDA-Stanford Competitive Progra... CV8 in Unsupervise... Red Hat Academy System Design Primer
Pretty-print
{
  "container": "a71f4d704f08",
  "message": "Backend container is running"
}
```

```
Command Prompt
D:\scaling-demo>cd nginx
D:\scaling-demo\nnginx>type nginx.conf
events {}

http {

    upstream backend_servers {
        server backend:8080;
    }

    server {
        listen 80;

        location / {
            proxy_pass http://backend_servers;
        }
    }
}
```